Mini Lecture: European vs American Option Pricing in Python

Daron Baltazar

1. Core Idea

In a binomial model, the stock price can move **up** or **down** at each time step. Once the stock price tree is built, the option is priced by working **backward** through the tree.

The key difference between **European** and **American** options is:

- **European option:** can only be exercised at maturity
- **American option:** can be exercised early at any node

2. European Option Logic

For a European option, the value at each node is just the **discounted expected continuation value**:

$$V = e^{-r\Delta t}\left[pV_{\text{up}} + (1-p)V_{\text{down}}\right]$$

This means the model assumes the option holder always waits until maturity.

Code idea:

```
V[j, i] = hold
```

3. American Option Logic

For an American option, the model must check two possibilities at each node:

1. **Hold the option** (continuation value)
2. **Exercise now** (intrinsic value)

So the value becomes:

$$V = \max(\text{hold}, \text{exercise})$$

For a call:

$$\text{exercise} = \max(S-K, 0)$$

For a put:

$$\text{exercise} = \max(K-S, 0)$$

Code idea:

```
V[j, i] = max(hold, exercise)
```

4. How This Fits Into the Code

Here is the backward induction structure:

```
for i in range(N - 1, -1, -1):
    for j in range(i + 1):
        hold = np.exp(-r * dt) * (
            p * V[j + 1, i + 1] + (1 - p) * V[j, i + 1]
        )

        if option_type == "call":
            exercise = max(S[j, i] - K, 0)
        else:
            exercise = max(K - S[j, i], 0)

        if style == "american":
            V[j, i] = max(hold, exercise)
        else:
            V[j, i] = hold
```

5. What Changed From European to American?

Only one main thing changes:

- **European:** use only the continuation value
- **American:** compare continuation value to immediate exercise

So in your code, the American version is basically:

$$\text{European backward induction} + \text{early exercise check}$$

6. Financial Intuition

- **European call:** must wait until expiration
- **American call:** may exercise early, but for non-dividend stocks this is usually not optimal
- **American put:** early exercise can be valuable, so it is often worth more than a European put

7. Practice Prompts

Prompt 1: Change the code from `style = "european"` to `style = "american"`. What changes in the final price?

Prompt 2: Why does the American model require `max(hold, exercise)`?

Prompt 3: Would you expect an American put to be worth more, less, or the same as a European put? Why?

8. Key Takeaway

The binomial tree makes American option pricing possible because it checks the value of exercising **at every node**, not just at maturity.

$$\textit{American option pricing} = \textit{European backward induction} + \textit{early exercise check}$$